

User-Level Coordinated Spread and Intervention in Misinformation Networks

Semi-Supervised Learning and LLM-Assisted Labelling Framework

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What is State of the Art?

State of the Art (SOTA) refers to the highest-performing and most advanced methods currently available in a research domain.

- What SOTA means: highest performing methods on standard benchmarks at this point in time.
- Requires evaluation on PHEME, CoAID. (SemEval-2019 Task 7 - Currently reviewing)
- Must outperform all prior transformer-GNN hybrid methods published at ACL/EMNLP/WWW/AAAI 2022–2025.
- A system is not SOTA by design alone — only by demonstrated benchmark results.

In This Research

Our project studies modern misinformation detection using transformer encoders, coordination analysis, semi-supervised learning, and LLM-assisted labelling.

Impact on the Research Pipeline

- Without SOTA architecture, the 79.1% BERT baseline has no path to improvement.
- BERTweet tokenization directly addresses the Twitter vocabulary gap that limits BERT.
- DeBERTa-v3 disentangled attention addresses the positional sensitivity gap in short tweet text.
- Llama 3 rationale features address the cross-domain generalization gap.
- SSL pipeline addresses the annotation bottleneck that prevents scaling beyond CoAID.

Research Impact

My project extends beyond traditional fake-news classification by combining: coordination analysis, SSL-based labelling, transformer encoders, and LLM-driven pseudo annotation into a unified pipeline.

Core Novel Contributions

- User-level coordination analysis instead of isolated tweet analysis.
- Scaling 2,907 labels to 35,000–70,000 without human annotation and Three-stage semi-supervised labelling pipeline.
- Semi-supervised learning for scalable annotation.
- Llama 3 assisted pseudo-labelling for ambiguous cases.
- DeBERTa-v3 used for contextual stance classification.
- BERTweet tokenizer optimized for Twitter/X text.

Key Innovation

The novelty lies in combining coordination behaviour modelling with advanced transformer-based text understanding and automated SSL-driven labelling.

Transformer Encoder Models

Model	Main Strength
BERT	Bidirectional contextual understanding
RoBERTa	Improved BERT pretraining efficiency
DistilBERT	Lightweight fast inference
ELECTRA	Efficient discriminative pretraining
XLNet	Permutation-based contextual learning
DeBERTa-v3	Strong contextual attention and disentangled embeddings
BERTweet	Optimized for Twitter/X social media text

Most Relevant Models

BERTweet and DeBERTa-v3 are selected because my project focuses on noisy, short-form social media misinformation content.

LLM Models Used in NLP Research

Model	Main Application
GPT Series	General reasoning and generation
Llama 3	Open-weight reasoning and pseudo-labelling
Mistral	Lightweight efficient inference
Gemma	Compact research-oriented model
Claude	Long-context reasoning
DeepSeek	Cost-efficient reasoning tasks

Project Usage

Llama 3-8B is used for pseudo-labelling and contextual reasoning over uncertain samples during the SSL workflow.

Why BERTweet?

- Pretrained directly on Twitter/X posts.
- Handles hashtags, mentions, emojis, and slang efficiently.
- Better tokenization for social-media-specific language.
- Improves semantic understanding of misinformation posts.
- Reduces information loss during preprocessing.
- BERTweet model itself is not fine-tuned or used for classification — only its tokenizer vocabulary is used as a preprocessing step before DeBERTa encoding.

Reason for Selection

Since the dataset originates from X/Twitter, BERTweet provides domain-specific language understanding unavailable in generic tokenizers.

Why DeBERTa-v3?

- Strong contextual attention mechanism.
- Uses disentangled attention representation.
- Higher classification accuracy than standard BERT.
- Performs well on misinformation and sentiment tasks.
- Better understanding of contextual contradiction means it handles negation and positional cues better due to disentangled attention mechanism.

Reason for Selection

DeBERTa-v3 is selected as the final classification encoder because of its superior contextual reasoning capability on complex misinformation text.

Combined Architecture Logic

- BERTweet specializes in social-media token representation.
- DeBERTa-v3 specializes in contextual classification.
- BERTweet provides Twitter-native tokenization as input.
- Then DeBERTa-v3 performs all contextual encoding on those tokens using disentangled attention.
- this may enhance misinformation detection accuracy which needs to be implemented to see the actual result.

Pipeline Integration

BERTweet improves token-level understanding of noisy X posts while DeBERTa-v3 performs high-quality contextual classification over the processed representations.

Role of Llama 3-8B

- Llama 3 serves two important roles,
 1. Offline rationale feature extractor for all samples
 2. Generates pseudo-labels for uncertain samples like low-confidence ambiguous samples (DeBERTa confidence 0.50–0.85).
- Performs contextual reasoning over misinformation narratives.
- Reduces manual annotation dependency.
- Useful for low-confidence classification cases.
- Supports scalable semi-supervised learning.

Project Contribution

Llama 3 acts as an intelligent pseudo-labelling assistant within the SSL pipeline to improve dataset scalability.

Why Semi-Supervised Labelling

- Only 2,907 manually labeled pairs available from the baseline study.
- Manual annotation does not scale to MC-Fake (~100K posts) or PHEME (6,425 threads).
- Semi-supervised learning expands the labeled set using the data's own structure and model confidence.
- Three sequential stages are applied — each handling what the previous stage could not.

Stage	Method	Who Does the Labelling
Stage A	Graph Propagation	Graph topology only — no model, no inference
Stage B	DeBERTa Self-Training	DeBERTa-v3 acts as teacher with BERTweet tokenization
Stage C	Llama 3 Reasoning	Llama 3-8B for ambiguous cases, confidence 0.50–0.85

Expected Outcome

From 2,907 seed labels to 35,000–70,000 effectively labeled instances. A 12–24× expansion with zero additional human annotation.

Core Principle and Operation

- Nodes structurally connected in the coordination graph tend to share the same label.
- Users in the same coordination cluster post similar content at similar times — their reply stances are likely similar.
- The existing C-Network graph from the baseline study is reused directly. No new data collection is required.
- Labels are diffused outward through graph edges proportional to edge weight and structural proximity.
- **No model is trained or run at this stage.** It is a pure graph algorithm.
- Each unlabeled node receives a soft probability distribution over Supporting, Neutral, and Intervening from its labeled neighbors.
- **Example:** A node connected to 3 Intervening and 1 Supporting labeled node receives soft label Intervening with probability 0.75.
- Nodes with maximum class probability ≥ 0.80 –0.85 are converted to hard pseudo-labels and added to the training set.
- Nodes below the threshold are passed to Stage B. Threshold is validated against the held-out portion of the 2,907 original labels.

Stage A Output

Produces 20,000–50,000 pseudo-labels depending on graph density.

Why This Stage Is Needed

- Stage A only labels nodes connected within the existing C-Network graph.
- Tweets from MC-Fake and PHEME have no graph edges to CoAID nodes — graph propagation cannot reach them.
- Disconnected nodes within CoAID itself also remain unlabeled after Stage A.
- For these tweets, the only available labelling signal is text content — requiring a text model.
- BERTweet tokenization is applied to all input text before every round of DeBERTa inference.

The Iterative Self-Training Loop

- **Round 1:** DeBERTa-v3 fine-tuned on 2,907 original labels + Stage A pseudo-labels. Run on all unlabeled MC-Fake and PHEME replies. Predictions with confidence ≥ 0.85 accepted as new pseudo-labels.
- **Round 2:** DeBERTa retrained on the expanded set from Round 1. Run again on remaining unlabeled data. More high-confidence predictions added.
- **Rounds 3–5:** Repeats until convergence — defined as fewer than 500 new pseudo-labels added per round. Each round adds 8,000–15,000 labels.

Stage B Output

Adds 10,000–15,000 pseudo-labels for nodes that graph propagation could not reach, drawn from CoAID, MC-Fake, and PHEME.

Why This Stage Is Required

A residual set of tweets remains after Stages A and B — cases that neither graph topology nor DeBERTa confidence can resolve:

- Replies using heavy sarcasm or irony where surface text contradicts actual stance.
- Implicit stance expressed through what is quoted or linked rather than directly stated.
- Identified by DeBERTa's maximum class probability falling between 0.50 and 0.85 — genuinely uncertain, not confidently wrong.

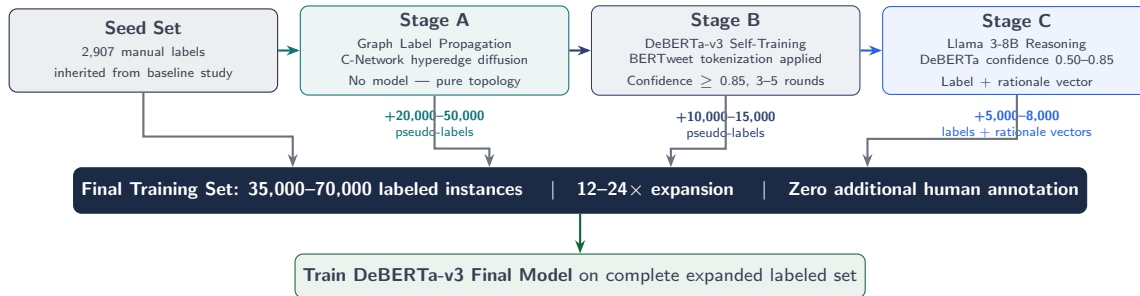
How Llama 3-8B Labels These Cases

"The parent tweet contains COVID-19 misinformation. Does the reply (a) Support, (b) remain Neutral, or (c) Intervene against it? Think step by step considering tone, claims made, and the relationship between the two tweets. Provide reasoning then give a single final label."

- The returned label is recorded as the pseudo-label for that tweet.
- The returned reasoning text is encoded into 256-dimensional **rationale vector** stored as additional feature.
- **Dual output:** one Llama 3 call produces both a label and an interpretable reasoning feature.

Stage C Output

Adds 5,000–8,000 pseudo-labels for residual ambiguous cases, each accompanied by a 256-dimensional rationale feature vector.



Stage A

C-Network hyperedge structure propagates known labels to connected unlabeled nodes. No model training or inference required.

Stage B

DeBERTa-v3 with BERTweet tokenization labels disconnected nodes in CoAID, MC-Fake, and PHEME iteratively.

Stage C

Llama 3-8B for ambiguous cases only. Produces a pseudo-label and a 256-dim rationale feature vector per sample.